WORKED SOLUTION

STEP I 2011, Question 3

Pure • Trigonometric Identities

STEP I 2011, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q3

The key idea

The product of two sines at $\frac{1}{3}\pi \mp \theta$ collapses neatly by a difference-of-squares trick, which proves $(*)$ in a line. After that, the cotangent sums come from differentiating a product: the derivative of a product of three factors, divided by the product itself, is a sum of three "logarithmic derivatives", and for sines those are cotangents. Doing the same to the cosine version gives tangents, and $\tan x + \cot x$ is a cosecant.

Solution

The identity $(*)$

Write $a = \frac{1}{3}\pi$. By the compound-angle formulae,

$$\begin{aligned}\sin(a-\theta)\sin(a+\theta) &= (\sin a\cos\theta - \cos a\sin\theta)(\sin a\cos\theta + \cos a\sin\theta)\\ &= \sin^2 a\cos^2\theta - \cos^2 a\sin^2\theta\\ &= \sin^2 a\,(1-\sin^2\theta) - (1-\sin^2 a)\sin^2\theta = \sin^2 a - \sin^2\theta.\end{aligned}$$

With $\sin^2 a = \frac{3}{4}$,

$$4\sin\theta\sin(\tfrac{1}{3}\pi-\theta)\sin(\tfrac{1}{3}\pi+\theta) = 4\sin\theta\left(\tfrac{3}{4}-\sin^2\theta\right) = 3\sin\theta - 4\sin^3\theta = \sin 3\theta,$$

by the triple-angle formula. This proves $(*)$, for every θ.

Part (i)

Since $(*)$ holds for all θ, we may differentiate both sides. By the product rule,

$$\begin{aligned}4\Big[\cos\theta\sin(a-\theta)\sin(a+\theta) &- \sin\theta\cos(a-\theta)\sin(a+\theta)\\ &+ \sin\theta\sin(a-\theta)\cos(a+\theta)\Big] = 3\cos 3\theta,\end{aligned}$$

where the minus sign in the middle term comes from the chain rule applied to $a-\theta$.

Ask yourself How do we turn this into cotangents?

Divide both sides by $\sin 3\theta$, which by $(*)$ equals $4\sin\theta\sin(a-\theta)\sin(a+\theta)$. Each term on the left loses all its sines except the one that was differentiated. This is allowed whenever $\sin 3\theta \neq 0$, and then all three sines are non-zero too. We get

$$\cot\theta - \cot(\tfrac{1}{3}\pi - \theta) + \cot(\tfrac{1}{3}\pi + \theta) = 3\cot 3\theta \qquad (\sin 3\theta \neq 0). \tag{A}$$

Now put $\theta = \frac{1}{9}\pi$, where $\sin 3\theta = \sin\frac{1}{3}\pi \neq 0$. Since $\frac{1}{3}\pi - \frac{1}{9}\pi = \frac{2}{9}\pi$ and $\frac{1}{3}\pi + \frac{1}{9}\pi = \frac{4}{9}\pi$,

$$\cot\tfrac{1}{9}\pi - \cot\tfrac{2}{9}\pi + \cot\tfrac{4}{9}\pi = 3\cot\tfrac{1}{3}\pi = \frac{3}{\sqrt{3}} = \sqrt{3},$$

as required.

Part (ii)

Put $\theta = \frac{1}{6}\pi - \phi$ in $(*)$. The four sines become

$$\begin{aligned}
\sin\theta = \sin(\tfrac{1}{6}\pi - \phi) &= \cos(\tfrac{1}{3}\pi + \phi),\\
\sin(\tfrac{1}{3}\pi - \theta) = \sin(\tfrac{1}{6}\pi + \phi) &= \cos(\tfrac{1}{3}\pi - \phi),\\
\sin(\tfrac{1}{3}\pi + \theta) = \sin(\tfrac{1}{2}\pi - \phi) &= \cos\phi,\\
\sin 3\theta = \sin(\tfrac{1}{2}\pi - 3\phi) &= \cos 3\phi,
\end{aligned}$$

using $\sin(\frac{1}{2}\pi - x) = \cos x$ each time. So, renaming ϕ as θ,

$$4\cos\theta\cos(\tfrac{1}{3}\pi - \theta)\cos(\tfrac{1}{3}\pi + \theta) = \cos 3\theta \qquad \text{for all } \theta. \tag{**}$$

Dividing $(*)$ by $(**)$, the factors of 4 cancel and each sine over cosine is a tangent; taking reciprocals,

$$\cot\theta\cot(\tfrac{1}{3}\pi - \theta)\cot(\tfrac{1}{3}\pi + \theta) = \cot 3\theta,$$

valid wherever both sides are defined, that is, where $\sin 3\theta \neq 0$ and $\cos 3\theta \neq 0$ (each of these makes the three corresponding factors non-zero).

For the cosecant sum, we differentiate $(**)$ just as in part (i):

$$\begin{aligned}
4\Big[&-\sin\theta\cos(a-\theta)\cos(a+\theta) + \cos\theta\sin(a-\theta)\cos(a+\theta)\\
&\qquad - \cos\theta\cos(a-\theta)\sin(a+\theta)\Big] = -3\sin 3\theta.
\end{aligned}$$

Dividing by $\cos 3\theta = 4\cos\theta\cos(a-\theta)\cos(a+\theta)$ (when it is non-zero) and multiplying by -1,

$$\tan\theta - \tan(\tfrac{1}{3}\pi - \theta) + \tan(\tfrac{1}{3}\pi + \theta) = 3\tan 3\theta. \tag{B}$$

The signs in (A) and (B) match, so we can add them.

Ask yourself Why would adding tangents and cotangents help?

Because $\tan x + \cot x$ is a single cosecant:

$$\tan x + \cot x = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \frac{2}{\sin 2x} = 2\operatorname{cosec} 2x.$$

So adding (A) and (B), whenever $\sin 6\theta \neq 0$, gives

$$2\operatorname{cosec} 2\theta - 2\operatorname{cosec}(\tfrac{2}{3}\pi - 2\theta) + 2\operatorname{cosec}(\tfrac{2}{3}\pi + 2\theta) = 6\operatorname{cosec} 6\theta.$$

Now take $\theta = \frac{1}{18}\pi$, so that $2\theta = \frac{1}{9}\pi$ and $6\theta = \frac{1}{3}\pi$ (with $\sin 6\theta \neq 0$). Since $\frac{2}{3}\pi - \frac{1}{9}\pi = \frac{5}{9}\pi$ and $\frac{2}{3}\pi + \frac{1}{9}\pi = \frac{7}{9}\pi$, dividing by 2 gives

$$\operatorname{cosec} \tfrac{1}{9}\pi - \operatorname{cosec} \tfrac{5}{9}\pi + \operatorname{cosec} \tfrac{7}{9}\pi = 3\operatorname{cosec} \tfrac{1}{3}\pi = 3 \cdot \frac{2}{\sqrt{3}} = 2\sqrt{3},$$

as required.

Checks

- At $\theta = \frac{1}{6}\pi$, $(*)$ reads $4 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot 1 = 1 = \sin \frac{1}{2}\pi$, and $(**)$ at $\theta = 0$ reads $4 \cdot 1 \cdot \frac{1}{2} \cdot \frac{1}{2} = 1 = \cos 0$.
- Numerically, $\cot 20^\circ - \cot 40^\circ + \cot 80^\circ \approx 2.7475 - 1.1918 + 0.1763 = 1.7321 \approx \sqrt{3}$, and $\operatorname{cosec} 20^\circ - \operatorname{cosec} 100^\circ + \operatorname{cosec} 140^\circ \approx 2.9238 - 1.0154 + 1.5557 = 3.4641 \approx 2\sqrt{3}$.
- Identities (A), (B) and the cotangent product can be tested at any random θ on a calculator; we checked them at a few thousand random values.

Common slips

- Forgetting the chain-rule minus sign when differentiating $\sin(\frac{1}{3}\pi - \theta)$, which spoils the alternating signs in (A).
- Dividing by $\sin 3\theta$ without noting that it must be non-zero at the value substituted.
- Trying to get the cosecant sum from the cotangent *product*: it is the *sums* (A) and (B) that combine.

Beyond STEP

The trick of dividing a derivative by the function is logarithmic differentiation: $\frac{\mathrm{d}}{\mathrm{d}\theta} \ln|f| = f'/f$ turns products into sums. Pushed further, sin itself can be written as an infinite product over its zeros, and logarithmic differentiation then gives the famous partial-fraction expansion $\pi \cot \pi x = \sum_n \frac{1}{x-n}$; identity $(*)$ is a finite shadow of this. These ideas are developed in complex analysis; Needham's *Visual Complex Analysis* is a beautiful place to start.

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